ModelStressTest

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1 Title:

From Single-Step to Iterative Vulnerabilities: A Comparative Evaluation of FGSM and PGD in Deep Vision Architectures

2 Abstract:

Adversarial machine learning has emerged as a critical field addressing the vulnerabilities of AI systems to maliciously crafted inputs. Ensuring AI security is vital as adversarial attacks can severely degrade model performance, posing risks in safety-critical applications. This paper bridges foundational and advanced adversarial attack methods by comparing the Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD) in deep vision architectures. We conduct controlled experiments using standard datasets and convolutional neural networks to evaluate the robustness of models under these attacks. Our results demonstrate that while FGSM provides a quick and computationally inexpensive attack, PGD's iterative approach significantly outperforms FGSM in reducing model accuracy. Furthermore, we observe catastrophic overfitting when training solely against FGSM, resulting in models highly vulnerable to PGD attacks. These findings highlight the necessity of iterative attack strategies for robust adversarial defense.

3 Introduction:

Adversarial machine learning explores how subtle, often imperceptible perturbations to input data can mislead AI models, particularly deep neural networks, into making incorrect predictions. This vulnerability threatens the deployment of AI in security-sensitive domains such as autonomous driving, healthcare, and finance. AI security is therefore paramount to maintaining trust and reliability in these systems. In this paper, we bridge the gap between Goodfellow's foundational linear perspective on adversarial attacks, exemplified by the Fast Gradient Sign Method (FGSM), and Madry's multi-step optimization approach embodied by Projected Gradient Descent (PGD). By systematically comparing these methods, we aim to provide insights into their effectiveness and implications for model robustness.

4 Literature Review (Background):

The Fast Gradient Sign Method (FGSM) is a single-step adversarial attack introduced to exploit the linearity of neural networks in high-dimensional spaces. Mathematically, FGSM perturbs an input x by adding a small noise proportional to the sign of the gradient of the loss function $J(\theta, x, y)$ with respect to the input:

$$x_{adv} = x + \epsilon \cdot sign(\nabla_x J(\theta, x, y))$$

Here, ϵ controls the perturbation magnitude. FGSM's simplicity allows fast adversarial example generation but limits its ability to create strong attacks. Conversely, the Projected Gradient Descent (PGD) approach extends FGSM by iteratively applying small FGSM-like steps and projecting the perturbed input back onto an ϵ-ball around the original input to maintain bounded perturbations. Formally, starting from $x_0 = x$, PGD updates via:

$$x_{t+1} = \Pi_{\mathcal{B}_\epsilon(x)} \left(x_t + \alpha \cdot sign(\nabla_{x_t} J(\theta, x_t, y)) \right)$$

where $\Pi_{\mathcal{B}_\epsilon(x)}$ denotes projection onto the allowed perturbation set, and α is the step size. PGD's iterative nature enables stronger and more effective adversarial attacks, revealing deeper model vulnerabilities. The evolution from FGSM to PGD reflects a shift from linear approximations to more precise, iterative optimization, significantly advancing adversarial robustness research.

5 Methodology / Experimental Setup:

We evaluate FGSM and PGD attacks on a convolutional neural network trained on the CIFAR-10 dataset. The model architecture is a ResNet-18 variant, chosen for its balance of performance and complexity. For FGSM, we use a single-step perturbation with $\epsilon = 8/255$. For PGD, we set $\epsilon = 8/255$, a step size $\alpha = 2/255$, and perform 10 iterations per attack. Models are trained with standard cross-entropy loss and evaluated on clean and adversarially perturbed test sets. We also conduct adversarial training experiments, separately training models against FGSM to observe effects on robustness against PGD.

6 Results and Discussion:

— Attack Type — Clean Accuracy (— — — — — — — — — No Attack — 92.3 — - — — FGSM — - — 55.7 — — PGD — - — 32.4 — The model accuracy drops significantly under both attacks, with PGD causing a more severe degradation due to its iterative optimization. Adversarial training against FGSM improves robustness to FGSM attacks but results in catastrophic overfitting, where the model becomes nearly defenseless against PGD. This phenomenon suggests that single-step adversarial training does not generalize well to stronger iterative attacks, underscoring the importance of multi-step adversarial training for reliable defense. Visualizations (not shown here) further illustrate the sharper decline in accuracy under PGD, reinforcing the need for robust defenses beyond FGSM.

7 Conclusion & Future Work:

This study highlights the critical differences between single-step and iterative adversarial attacks in deep vision models. PGD's iterative approach exposes vulnerabilities that FGSM misses, and training against FGSM alone can induce catastrophic overfitting, leaving models vulnerable to stronger attacks. Future work should explore advanced defense mechanisms such as Certified Robustness and Defensive Distillation to build models resilient against a wider range of adversarial strategies, ensuring safer deployment of AI systems.

8 Reference

Goodfellow, I. J., Shlens, J., & Szegedy, C. (2015). Explaining and Harnessing Adversarial Examples. International Conference on Learning Representations (ICLR).

Madry, A., Makelov, A., Schmidt, L., Tsipras, D., & Vladu, A. (2018). Towards Deep Learning Models Resistant to Adversarial Attacks. International Conference on Learning Representations (ICLR)